HAM: Hierarchical Attention Mamba With Spatial–Frequency Fusion for Remote Sensing Image Super-Resolution

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***Abstract*—Vision Mamba shows significant potential for enhancing remote sensing image super-resolution (RSISR) performance due to its linear-complexity global modeling capabilities. However, existing Mamba-based RSISR models face two key limitations in handling remote sensing imagery. First, insufficient hierarchical feature interaction hinders the capture of multiscale spatial structures, which are crucial for reconstructing diverse patterns and fine textures. Second, inadequate local–global correlation extraction weakens the restoration of high-frequency details in conjunction with broader contextual information, ultimately limiting spatial consistency and image fidelity. To address these issues, we propose the hierarchical attention Mamba (HAM) network with spatial–frequency feature enhancement for RSISR. Key innovations include the hierarchical aggregation attention (HAA) module, which enhances the propagation and utilization of multilevel features, and the spatial–frequency information interaction module (SFIIM), which facilitates the fusion of spatial–frequency features. The SFIIM employs the dynamic frequency spectrum enhancement (DFSE) technique for discriminative feature learning through Fourier amplitude–phase decoupling, along with the hybrid spatial feature aggregator (HSFA), which utilizes a spatial-domain dual-branch channel-wise gating mechanism. Together, these techniques significantly enhance the model's ability for synchronous local–global feature extraction. Extensive evaluations across six RSISR benchmarks (including AID, DIOR, and DOTA) demonstrate that HAM achieves state-of-the-art performance, delivering superior image reconstruction quality while maintaining computational efficiency.**

***Index Terms*—Hierarchical aggregation attention (HAA), Mamba, remote sensing image (RSI), spatial–frequency fusion, super-resolution (SR).**

I. Introduction

THE acquisition of high-resolution (HR) remote sensing images (RSIs) is of great significance for implementing high-precision ground observation and image analysis [1], [2], [3]. Due to limitations in imaging technology and cost constraints, many existing RSIs exhibit low-resolution (LR). RSI super-resolution (RSISR) technology aims to reconstruct HR images from LR inputs, making it a crucial research area with broad applications in both academia and industry.

Recently, deep learning has been extensively applied to image super-resolution (SR) [4], [5], [6]. Initially, convolutional neural networks (CNNs) achieved remarkable success, with notable models such as the enhanced deep super-resolution network (EDSR) [7] and the residual dense network (RDN) [8]. Subsequently, more sophisticated attention mechanisms were integrated into CNNs, including the residual channel attention network (RCAN) [9] and the hybrid attention network (HAN) [10]. These CNN-based methods effectively capture local details, but the intrinsic spatial limitations of convolutional kernels restrict the receptive field, making it challenging to model RSIs with extensive spatial context.

However, vision transformers (ViTs) [11] employ self-attention (SA) mechanisms to enable global information exchange across all tokens in the input image [12], thereby achieving a global receptive field. However, this comes at the cost of quadratic computational complexity. To mitigate this computational burden, recent studies have introduced various strategies [13], [14], including window-based SA in SwinIR [15] and the adaptive sparse transformer (ART) [16], to achieve a balance between efficiency and performance.

In recent years, state space models (SSMs) [21], [22] have emerged as a promising paradigm in sequence modeling, offering a balance between computational efficiency and exceptional long-range dependency modeling. With its transformer-like capability for capturing long-range dependencies and linear computational complexity, Mamba represents a compelling choice for various visual tasks, including image restoration [19], [23], [24], deraining [25], shadow removal [26], and low-light enhancement [27], as well as image SR [28].

The development of RSISR in deep learning has closely paralleled progress in natural image SR. Early CNN-based methods, such as residual [29] and recursive networks [30], improved spatial detail recovery but lacked robustness to scene diversity, structural complexity, and scale variation, and could not capture global dependencies. To address these issues, attention mechanisms were introduced to guide CNNs toward salient features. For instance, Zhang et al. [31] designed a

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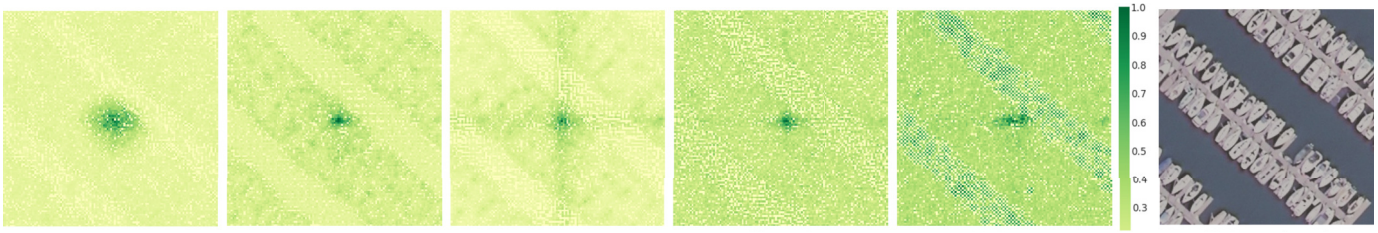


Fig. 1. Effective receptive field (ERF) visualization [17], [18] for HAN [10], ART [16], MambaIR [19], FMSR [20], and our proposed HAM model is presented. A broader distribution of dark areas indicates a larger ERF. As shown in (e), the water region, representing the global receptive field, appears darker compared to other models, suggesting stronger global feature capture. Furthermore, the vessel area, representing the local receptive field, shows selective darkening at various vessel positions, highlighting HAM's capability to effectively extract both global and local information. (a) HAN [10]. (b) ART [16]. (c) MambaIR [19]. (d) FMSR [20]. (e) Ours. (f) Ground-truth.

multiscale attention network, while Qiu et al. [32] proposed an edge-guided attention framework to enhance structural fidelity. Transformer-based models, leveraging SA for global context modeling, have gained wide adoption in RSISR. Xiao et al. [33] introduced a top-k token selective transformer, and Lei et al. [34] fused multilevel features to boost representation capability. Recently, SSM-based architectures have shown promise. Xiao et al. [20] proposed FMSR, a frequency-assisted Mamba-based network, while Zhi et al. [35] presented MambaFormerSR, which combines Mamba with transformer and uses a convolutional Fourier-based feed-forward module. Despite these efforts, Mamba's potential in RSISR remains underexplored.

Specifically, it is crucial to acknowledge the substantial differences between SR tasks for natural images and RSISR. Compared to natural images, remotely sensed images exhibit greater complexity, higher scene diversity, significant scale variations, and pronounced directional differences in feature characteristics. As a result, existing Mamba-based RSISR approaches [20], [35] still encounter the following challenges.

1) Inadequate Hierarchical Feature Interaction: Most SR networks currently adopt a hierarchical feature extraction strategy with unidirectional propagation, where initial layers acquire local textures while deeper layers obtain global semantic structures. This leads to a lack of interlayer feature interactions. While residual connections help mitigate low-level feature degradation to some extent, the simplistic fusion of shallow and deep features remains insufficient for adaptive, context-aware multilevel representation learning.

2) Insufficient Local–Global Synchronous Modeling: Visual Mamba processes image patches along fixed scanning trajectories, which introduce artificial causal dependencies and enforce uniform treatment of all spatial relationships. The attenuation of information in long-range token interactions constrains the adaptive modeling of noncausal local structures, thereby hindering the capture of macroscopic spatial context.

Based on the above background, we propose a novel Mamba-based RSISR model: hierarchical attention Mamba (HAM) with spatial–frequency fusion network. The core ideas of HAM can be described as follows. First, we introduce the hierarchical aggregation attention (HAA) mechanism, which dynamically assigns weights for adaptive interaction fusion of multiscale cross-layer features. Second, we develop the spatial–frequency information interaction module (SFIIM) to enhance spatial–frequency features and promote cross-domain fusion. This module consists of a dynamic frequency spectrum enhancement (DFSE), which employs discriminative feature learning through Fourier amplitude–phase decoupling, and a hybrid spatial feature aggregator (HSFA) that utilizes a spatial-domain dual-branch channel-wise gating mechanism. These techniques significantly enhance the model's capability for synchronous local–global feature extraction.

Fig. 1 illustrates a comparison of the ERF visualizations, showcasing the global and local modeling capabilities of several state-of-the-art SR methods based on CNN, transformers, and Mamba. The results in Fig. 1 indicate that: although the CNN-based method HAN [10] in Fig. 1(a) increases the local receptive field through certain optimization techniques, its global modeling ability remains the weakest (as indicated by the large light-colored area). In contrast, the transformer-based method ART [16] in Fig. 1(b) and the Mamba-based methods (MambaIR [19], FMSR [20], and our method) in Fig. 1(c)–(e) demonstrate stronger global modeling capabilities (more areas are darker in color). It is important to note that MambaIR in Fig. 1(c) represents a four-way global receptive field, defined by horizontal and vertical scanning directions, while FMSR in Fig. 1(d) represents a global receptive field achieved through basic frequency assistance. Whereas our proposed method in Fig. 1(e) leverages the global receptive field with enhanced depth, demonstrating significantly superior local and global modeling capabilities.

To further investigate the impact of Fourier-transformed magnitude and phase components on image SR, we perform a component-swapping experiment: the magnitude and phase of the HR image and the LR image (after bicubic interpolation) are exchanged to generate two synthetic variants: HR magnitude + LR phase (high amplitude and low phase) and HR phase + LR magnitude (high phase and low amplitude), as illustrated in Fig. 2. As shown in Fig. 2(c), the image with high amplitude and low phase exhibits overall sharpness but fails to reconstruct smooth yellow arcs or small ground cracks accurately. This suggests that amplitude primarily influences global image clarity, but may cause distortion in fine details. In contrast, Fig. 2(d), (high phase and low amplitude) appears blurrier globally, yet the yellow arcs and white edges are better defined than in the upsampled LR image, with superior

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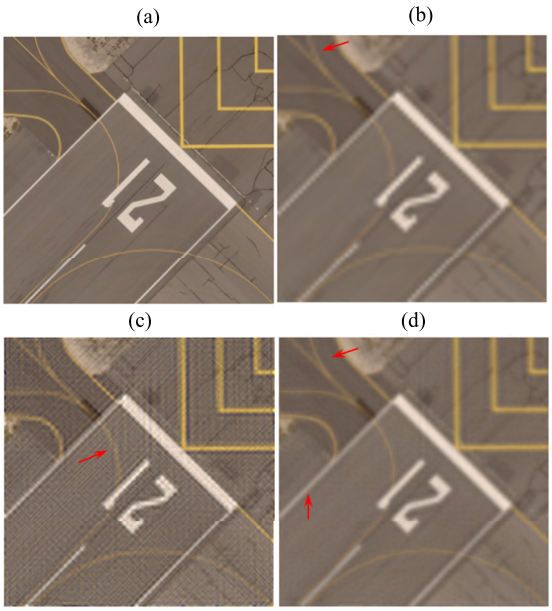


Fig. 2. Visual comparison by decoupling amplitude and phase in the Fourier transform domain. (a) HR. (b) LR (upsampled). (c) HRmag + LRphase. (d) HRphase + LRmag.

fine-detail reproduction compared to Fig. 2(c). This suggests that phase plays a more critical role in texture and structural detail recovery. Notably, reconstructing images using either HR amplitude or phase alone yields suboptimal results. In particular, as illustrated in Fig. 2(c), amplitude–phase mismatch can result in noticeable texture artifacts. These findings validate that decoupling the amplitude and phase components and learning them separately can effectively enhance image SR performance.

Our principal contributions are as follows.

1) We propose the HAA mechanism to efficiently improve the propagation and utilization of multiscale features extracted by multilevel networks.
2) We develop the SFIIM to promote spatial–frequency feature fusion and enhance the model's capability for synchronous local–global feature extraction.
3) Comprehensive evaluations on six benchmark RSI datasets demonstrate the superior performance of our proposed HAM in balancing image reconstruction quality and efficiency.

II. Related Work

A. State Space Models

SSMs [21], [36], originally derived from classical control theory, have recently been introduced into deep learning, garnering widespread attention due to their exceptional ability to model long-range dependencies. Gu et al. [21] first introduced the structured state space sequence (S4) model to model long-range dependencies, based on the concept of linear sequence scaling, which has sparked extensive research. Subsequently, Smith et al. [36] proposed S5, which retains the computational efficiency of S4 while introducing a parallel scanning mechanism to further enhance performance, leading to significant advancements in natural language processing (NLP) tasks. Recently, Gu and Dao [37] introduced Mamba, an extension of the S4 architecture that integrates a data-dependent selective mechanism and a hardware-efficient design. These enhancements streamline the computational flow and architecture, endowing Mamba with linear complexity while achieving transformer-like expressive capability.

B. Visual Mamba

SSMs exhibit linear computational complexity and efficiently capture global features, providing significant advantages in tasks such as image processing. In the visual domain, Zhu et al. [38] introduced the ViM model, inspired by the ViT. This model constructs Mamba blocks with bidirectional paths and integrates them into the ViM network after embedding positional information into the image patches. Liu et al. [39] proposed the VMamba model, which adopts a cross-scanning approach by scanning the image in four directions and introduces a 2-D selective scanning (SS2D) module. MambaIR [19] utilized the VMamba architecture to explore the potential of Mamba in image restoration tasks, establishing a Mamba-based benchmark model for image restoration. Liu et al. [39] proposed the VMambaIR model based on the U-Net network, in which the omnidirectional selection module (OSS) models in six directions. In the field of RSISR, Xiao et al. [20] proposed the frequency-assisted Mamba model (FMSR), based on MambaIR. The MambaFormerSR proposed by Zhi et al. [35] is the first RSISR network to combine Mamba with the transformer architecture. While these methods demonstrate the potential of SSMs, they suffer from several limitations, including fixed scanning orders that induce directional bias in aerial image reconstruction, shallow feature degradation during deep propagation, and inadequate exploitation of the synergies between spatial and frequency domains—issues that hinder the model's ability to effectively capture complex image details.

C. Frequency Learning

The Fourier transform is a widely used technique for extracting frequency information in signal analysis. It provides valuable global statistical information about the signal and effectively captures long-range dependencies. In light of this, Fourier transform methods have been incorporated into the image restoration field in recent years. Sinha et al. [40] and Zhang et al. [41], respectively, proposed networks utilizing nonlocal fast Fourier convolution [42] to enhance SR by expanding the receptive field and capturing long-range dependencies. The spatial–frequency hybrid network (SFHN) proposed by Wu et al. [43] performs complementary learning of information extracted from both the frequency and spatial domains.

In addition, FMambaIR, proposed by Luan et al. [44] within the Mamba network framework, adopts a dual-branch global modeling strategy that integrates spatial-domain Mamba with Fourier transform domain attention, effectively enhancing underwater images. FreqMamba [45] analyzes Mamba from both spatial and frequency perspectives, incorporating the Fourier transform for global degradation modeling. DFSSM

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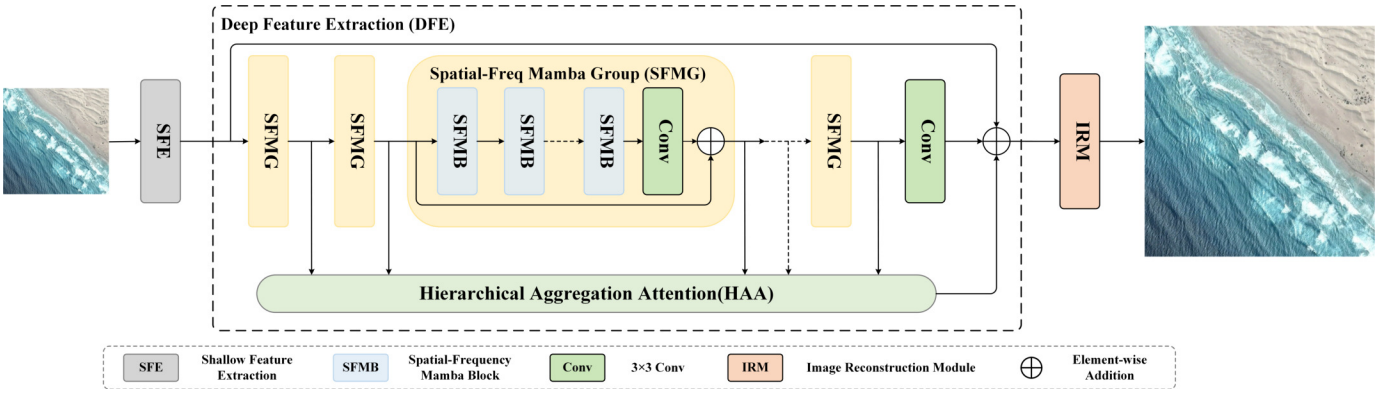


Fig. 3. Overview of the proposed HAM.

[25] enhances the visual state space module (VSSM) using a fast Fourier transform (FFT) feature extraction module, effectively removing rain streaks from high-frequency components. In the field of RSISR, FMSR [20] employs a frequency-assisted approach to help Mamba recover lost frequency details in RSIs.

Overall, these studies highlight the effectiveness of Fourier-domain methods in enhancing image details and improving quality. However, existing Mamba-based frequency learning methods in RSISR fail to fully leverage the Fourier transform's potential to improve SR by decoupling amplitude and phase.

III. Methodology

This section presents a comprehensive introduction to the proposed HAM Network with spatial–frequency fusion. We will outline the overall structure of HAM, followed by detailed explanations of each key component, including the spatial-frequency mamba block (SFMB) and its submodule SFIIM and HAA.

A. Overview of HAM

As illustrated in Fig. 3, the proposed HAM network architecture adheres to the classic residual in residual (RIR) framework, consisting of three core modules: the shallow feature extraction (SFE) module, the SFMG for hierarchical feature learning, and the image reconstruction module (IRM). The SFE module processes LR input image $I_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times C_{\mathrm{in}}}$, through a 3×3 convolutional layer, where C_{in} represents the input image's channel number, and H and W represent the image's height and width, respectively. This operation extracts shallow features $F_0 \in \mathbb{R}^{H\times W\times C}$, with C being the embedded channel dimension.

Subsequently, the shallow features F_0 are processed through stacked SFMG modules to generate hierarchical deep features

$$F_n = H_{\mathrm{SFMG}_n}\left(H_{\mathrm{SFMG}_{n-1}}\left(\cdots H_{\mathrm{SFMG}_1}\left(F_0\right)\right)\right) \tag{1}$$

where F_n represents the nth level deep features extracted by the nth SFMG, $n = 1, 2, \ldots, N$, where N is the number of SFMGs, and H_{SFMG_n} represents the nth SFMG used.

Each SFMG is composed of multiple stacked SFMBs, convolutional layers, and skip connections. For the nth SFMG, the input features F_{n-1} from the $(n-1)$th SFMG are first processed through the L stacked SFMBs to extract intermediate features $\hat{F}_L$

$$\hat{F}_L = H_{\mathrm{SFMB}_L}\left(H_{\mathrm{SFMB}_{L-1}}\left(\cdots H_{\mathrm{SFMB}_1}\left(F_{n-1}\right)\right)\right) \tag{2}$$

where $H_{\mathrm{SFMB}_L}(\cdot)$ denotes the Lth SFMB included in the SFMG.

Then, the intermediate features $\hat{F}_L$ are passed through a convolutional layer, with a skip connection, to obtain the intermediate deep features F_n of the nth SFMG. The convolutional layer follows the classic SR model, and the introduced skip connections implement an identity mapping, aggregating features from different layers and improving the stability of model training, i.e.,

$$F_n = \mathrm{Conv}\left(\hat{F}_L\right) + F_{n-1}. \tag{3}$$

To facilitate cross-layer feature interaction within the stacked SFMG architecture, we implement a multistage fusion strategy that aggregates hierarchical representations from all SFMGs through channel-wise Concatenation. This composite feature tensor is subsequently processed by the HAA module, which is formulated as

$$F_{\mathrm{HAA}} = H_{\mathrm{HAA}}\left(\mathrm{Concat}\left(F_1, F_2, \ldots, F_N\right)\right) \tag{4}$$

where F_{HAA} denotes the features generated by the HAA module, $H_{\mathrm{HAA}}(\cdot)$ represents the operation, Concat$(\cdot)$ denotes Concatenation operation, and N represents the number of SFMGs.

Finally, the HAA's features F_{HAA}, shallow features F_0, and the output features F_N of the final-stage SFMG are combined via element-wise addition and fed into the IRM to generate the SR result, which is formulated as

$$I_{\mathrm{SR}} = H_{\mathrm{IRM}}\left(F_0 + \mathrm{Conv}\left(F_N\right) + F_{\mathrm{HAA}}\right) \tag{5}$$

where $H_{\mathrm{IRM}}(\cdot)$ represents the IRM. As in previous works, the IRM consists of two 3×3 convolutional layers and a pixel-shuffle layer.

B. Spatial–Frequency Mamba Block

SFMB is the key component of SFMG, as shown in Fig. 4(b), and is primarily designed to learn global features from both the spatial and frequency domains. Specifically, the input features first undergo layer normalization (LN) and then use the VSSM [see Fig. 4(a)] to extract long-range dependency information in the spatial domain. The SS2D module in VSSM

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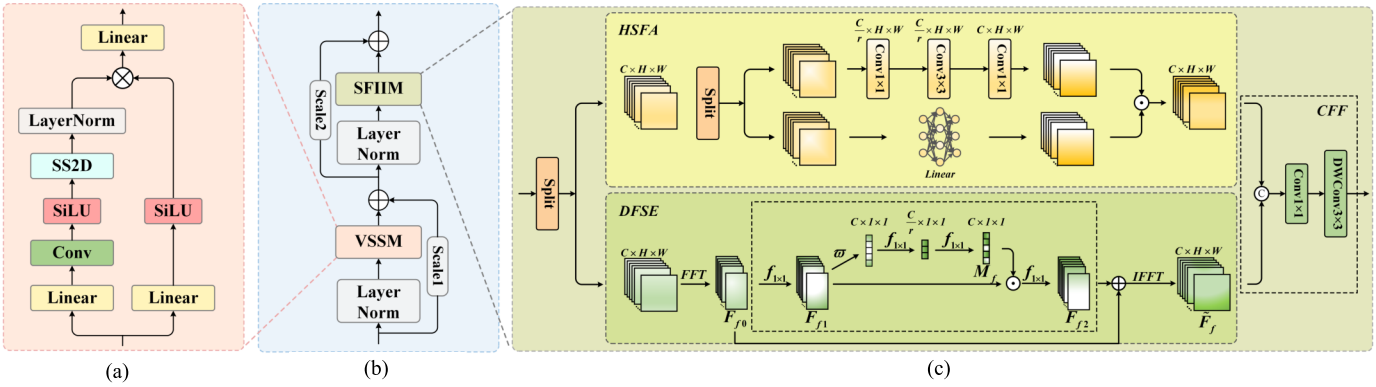


Fig. 4. Spatial–frequency Mamba blocks (SFMBs) are sequentially arranged within the spatial–frequency Mamba group (SFMG). Within SFMB, the VSSM, and the SFIIM work together for learning.

employs a cross-scanning strategy consistent with that used in MambaIR [19] and VMamba [39], scanning in four directions, top-left to bottom-right and bottom-right to top-left in the horizontal and vertical directions, respectively. A skip connection with a learnable scale factor of $\gamma_1 \in \mathbb{R}^c$ is also introduced. This design allows for more refined dynamic adjustments in the fusion of local texture features and global semantic features, thus enhancing the model's ability to recover details, especially in complex RSI scenarios. The first stage of the SFMB can be formulated as

$$x_j = \gamma_1 \times y_{j-1} + H_{\text{VSSM}}\left(\text{LN}\left(y_{j-1}\right)\right) \tag{6}$$

where y_{j-1} represents the output feature of the $(j-1)$th SFMB, x_j represents the intermediate feature of the jth SFMB, and $H_{\text{VSSM}}(\cdot)$ represents the VSSM module.

Subsequently, the output intermediate feature x_j is first normalized using LN, and then the SFIIM is used to fuse spatial and frequency domain features. Finally, another module with a learnable scale factor of $\gamma_2 \in \mathbb{R}^c$ adaptively aggregates the information from both the spatial and frequency domains, which can be expressed as

$$y_j = \gamma_2 \times x_j + H_{\text{SFIIM}}\left(\text{LN}\left(x_j\right)\right) \tag{7}$$

where y_j represents the output of the jth SFMB.

C. Spatial–Frequency Information Interaction Module

We designed a SFIIM, as shown in Fig. 4(c), consisting of a parallel HSFA unit, a DFSE unit, and a cross-domain feature fusion (CFF) unit for interaction fusion between the HSFA and DFSE.

1) Hybrid Spatial Feature Aggregator: To address the loss of local image correlations caused by the VSSM's 2-D to 1-D sequence scanning approach for modeling, we propose a HSFA unit that employs a dual-branch channel-wise gating structure, as illustrated in Fig. 4(c) (upper). One branch comprises two 1×1 convolutional layers for channel compression and recovery, along with a 3×3 convolutional layer for spatial feature extraction. The other branch utilizes a linear layer for interchannel learning. The output features are obtained by element-wise multiplication of the outputs from both branches. Given the input feature be $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, let F_{Conv} and F_{linear} denote the outputs of the convolutional and linear branches, respectively. The HSFA can be formulated as follows:

$$F_{\text{HSFA}} = F_{\text{Conv}} \odot F_{\text{linear}} \tag{8}$$

where $\odot$ denotes element-wise multiplication.

The element-wise multiplication of branch outputs maintains spatial structure consistency while introducing global context through weighted aggregation, achieving effective cross-spatial feature fusion and enhancing modeling ability.

2) Dynamic Frequency Spectrum Enhancement: Building on prior studies in image de-raining [25] and image restoration [44] tasks that analyzed the roles of amplitude and phase components in the Fourier domain—as well as our own analysis of these components in the SR task (see Section I)—we posit that phase features are primarily responsible for encoding local texture details, while amplitude features capture global structural clarity. To enhance the model's ability to capture both local and global correlations, we designed a DFSE unit, as illustrated in Fig. 4(c) (bottom). This component achieves adaptive improvement of global structural features and local texture details by decoupling amplitude features (which encode global structural correlations) and phase features (which capture local texture details) in the frequency domain, applying independent discriminative enhancement through channel attention (CA) mechanisms.

Specifically, the input feature $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ of the DFSE first undergoes FFT to obtain the spectrum representation $F_{f0} = \text{FFT}(F_{\text{in}})$. This output is then aggregated through pointwise convolution as follows:

$$F_{f1} = \sigma\left(f_{1\times 1}\left(F_{f0}\right)\right) \tag{9}$$

where $F_{f0} = \{A_f, P_f\}$ includes the amplitude (A_f) and phase (P_f) components derived from the Fourier transform, and $f_{1\times 1}(\cdot)$ represents 1×1 convolution operation applied to aggregate frequency-domain features.

Next, CA is applied separately to the amplitude and phase spectra to produce refined features F_{f2}. To dynamically modulate the importance of each frequency component and allow the model to adaptively select discriminative frequency bands for enhancement, we compute the channel weight mapping

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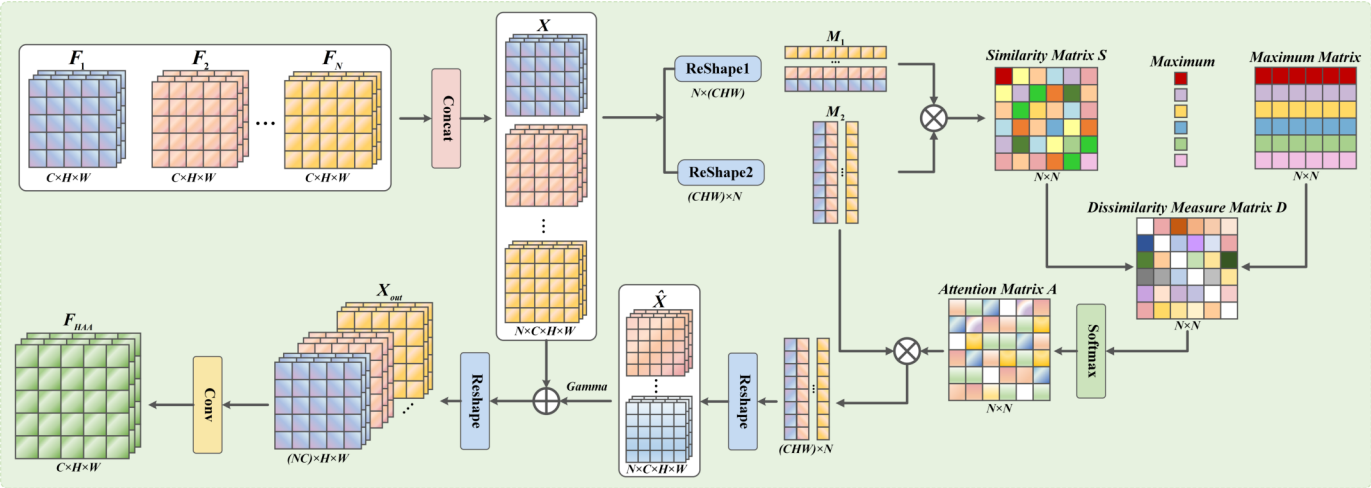


Fig. 5. Proposed HAA.

M_f from F_{f1} using 1×1 convolution and average pooling operations

$$M_f = \delta\left(f_{1\times1}\left(\sigma\left(f_{1\times1}\left(\varpi\left(F_{f1}\right)\right)\right)\right)\right) \tag{10}$$

where $\varpi(\cdot)$ represents avgpool operation.

Using the CA weights M_f, we further conduct discriminative learning on the amplitude and phase features in the frequency domain as follows:

$$F_{f2} = f_{1\times1}\left(M_f \cdot F_{f1}\right). \tag{11}$$

Finally, we fuse the original spectral domain features through a residual connection to obtain the enhanced frequency domain features, expressed by the following formula:

$$\tilde{F}_f = F_{f2} + F_{f0}. \tag{12}$$

The dynamically enhanced features are then reconstructed to the spatial domain using the inverse FFT (IFFT), maintaining coherent frequency-spatial information flow. The reconstruction process is formalized as follows:

$$F_{\text{DFSE}} = \text{IFFT}\left(\tilde{F}_f\right). \tag{13}$$

3) Cross-Domain Feature Fusion: After separately processing spatial and frequency information, cross-domain interaction is facilitated by the CFF unit. Specifically, the input feature $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ undergoes parallel processing in both domains. After spatial and frequency feature learning, the outputs are concatenated along the channel dimension to form a joint feature map encoding both spatial and frequency information. This concatenated map is then passed through a 1×1 convolutional layer, followed by a deep-wise 3×3 convolutional layer, to enable deep fusion of cross-domain features, as expressed in the following equation:

$$F_{\text{out}} = \text{CFF}\left(F_{\text{HSFA}}, F_{\text{DFSE}}\right). \tag{14}$$

D. Hierarchical Aggregation Attention

To better preserve low-level feature information and enhance the feature representation capability of the deep feature extraction network, we designed a HAA module to achieve interaction and adaptive discriminative fusion of multilevel and multiscale features.

As shown in Fig. 5, the HAA operates according to the following procedure. First, deep feature tensors from different hierarchical levels are Concatenated to create a feature matrix $X \in \mathbb{R}^{N\times C\times H\times W}$. This feature matrix is then reshaped to form the aggregated data matrix $M_1 \in \mathbb{R}^{N\times(\text{CHW})}$ and its transpose, $M_2 \in \mathbb{R}^{(\text{CHW})\times N}$, expressed as

$$\begin{aligned} M_1 &= \text{Reshape}\left(X\right) \in \mathbb{R}^{N\times(\text{CHW})} \\ M_2 &= M_1^T \in \mathbb{R}^{(\text{CHW})\times N} \end{aligned} \tag{15}$$

where $X = \text{Concat}(F_1, F_2, \ldots, F_N)$, and N represents the number of SFMGs used.

Next, the hierarchical feature similarity matrix S is computed by multiplying matrices M_1 and M_2. The row-wise maximum values of matrix S are derived, and S is subtracted from these maxima to generate a corresponding dissimilarity measure matrix D. These processes can be sequentially formulated as

$$\begin{aligned} S &= M_1 \otimes M_2 \in \mathbb{R}^{N\times N} \\ D &= \max\left(S\right) - S \in \mathbb{R}^{N\times N} \end{aligned} \tag{16}$$

where $\otimes$ represents matrix multiplication, and each element $S_{i,j}$ in the similarity matrix represents the correlation between the ith and jth layer features. $\max(\cdot)$ refers to taking the maximum value of each row and expanding it to the original size. The dissimilarity measure matrix D is designed to amplify interlayer feature discrepancies. By subtracting similarity scores from their row-wise maxima, we emphasize the most distinct features between layers, which naturally suppresses redundant correlations and enhances discriminative feature fusion.

The Softmax operation is applied to D to produce the attention score matrix $A \in \mathbb{R}^{N\times N}$. The hierarchical feature aggregation matrix M_2 is multiplied by A, followed by an inverse reshape operation. The resulting tensor $\hat{X}$ is multiplied by the learnable weight γ and summed with the original hierarchical feature aggregation tensor X through a residual

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connection. Finally, this output passes through a 1×1 convolutional layer to reduce channel dimensions, yielding the final output feature $F_{\text{HAA}} \in \mathbb{R}^{C\times H\times W}$. These processes can be sequentially formulated as

$$\begin{aligned} A &= \text{Softmax}\,(D) \in \mathbb{R}^{N\times N} \\ \hat{X} &= \text{Reshape}\,(M_2 \otimes A) \in \mathbb{R}^{N\times C\times H\times W} \\ X_{\text{out}} &= \text{Reshape}\left(\gamma\hat{X} + X\right) \in \mathbb{R}^{(NC)\times H\times W} \\ F_{\text{HAA}} &= \text{Conv}\,(X_{\text{out}}) \in \mathbb{R}^{C\times H\times W}. \end{aligned} \tag{17}$$

Here, we apply the Softmax function to the dissimilarity measure matrix D, resulting in a normalized attention weight matrix A. The attention matrix A assigns higher weights to larger differences and lower weights to smaller ones. This means that during aggregation, the model enhances the high-contribution feature layers while suppressing redundant ones. We introduce a residual connection and adopt a learnable weight γ to balance the influence of the original feature X and the aggregated feature $\hat{X}$, adaptively supplementing the low-level information that may be lost during the forward propagation process.

We adopt matrix D, defined as $D = \max(S)\text{-}S$ as in (16), as the basis for attention weights—diverging from conventional "maximum positive similarity" approaches. This design is motivated by two key principles. The first is the suppression of feature redundancy. Hierarchical networks often exhibit semantic redundancy in deeper layers. While similarity-based attention (e.g., softmax over S) reinforces dominant correlations, it may overlook unique information from under-represented layers. By inverting S, D prioritizes features with lower similarity, mitigating redundancy and enhancing feature diversity. The second is the enhancement of complementary fusion. Shallow and deep features capture fine-grained details and high-level semantics, respectively. Matrix D amplifies interlayer discrepancies, facilitating adaptive fusion of complementary information. This aligns with information-theoretic principles of negative selection, where suppressing correlated signals highlights novel patterns. The attention matrix $A = \text{Softmax}(D)$ as in (17) operationalizes these principles by: penalizing redundancy (assigning lower weights to high-similarity layer pairs); amplifying discriminative signals (assigning higher weights to distinct layers). This promotes the integration of diverse features across the network—such as combining low-level spatial textures with high-level semantic structures—ultimately enhancing the network's representation capacity.

To validate the proposed strategy, we further visualize HAA using local attribution maps (LAMs) [47]. As shown in Fig. 6(b), HAA attends to a broader range of spatially relevant pixels and aligns more closely with semantically meaningful regions, outperforming both the version without HAA [see Fig. 6(c)] and the variant in which HAA is replaced by squeeze-and-excitation (SE) attention [46] [see Fig. 6(d)]. They exhibit narrower activation, weaker discriminative capacity, and limited utilization of cross-layer disparities. This suggests that conventional attention mechanisms tend to reinforce redundant features while neglecting informative yet less prominent signals. These findings validate that the proposed D-driven attention mechanism effectively captures spatially diverse and semantically meaningful regions, which is essential for complex RSISR tasks. Additional supporting evidence is presented in subsequent experiments.

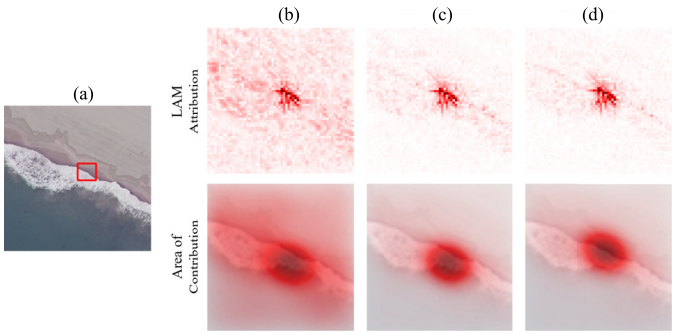


Fig. 6. Comparison of LAM results of using HAA and SE [46]. LAM [47] can visualize the SR network's dependence on the input pixels and the spatial range. SR networks with better performance often involve a larger range of input pixels. (a) HR image. (b) w HAA. (c) w/o HAA. (d) w SE layer.

IV. Experiments

In this section, we present a comprehensive evaluation of HAM by comparing its performance with several state-of-the-art image SR models. Furthermore, some ablation studies are conducted to systematically analyze the contribution of each component within HAM.

A. Model Details

For the HAM architecture, the deep feature extraction component consists of six SFMGs, each containing six SFMBs. This article focuses on $4\times$ SR. Following the network framework of MambaIR, we set the internal channel dimension to $C = 180$, and the hidden state size of the SSM is configured as $d_\text{state} = 16$ and the expansion ratio in the linear projection layer is set to $\lambda = 2$. We introduce two versions: HAM and HAM-Tiny. The only difference between these two versions lies in the channel dimension, which is 180 and 96 dimensions, respectively.

B. Training Details

Following prior studies [15], [16], [19], data augmentation techniques, including horizontal flipping and random rotations of 90°, 180°, and 270°, are applied to enhance the generalization capability of the model. During training, all SR models are retrained on the AID dataset using the L1 loss function. To facilitate effective learning, the original images are cropped into 64×64 patches. Our HAM is trained for 500 000 iterations using the Adam optimizer [59], with hyperparameters set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The initial learning rate is set at 2×10^{-4} and is decayed by a factor of 2 at predefined milestones to stabilize training and enhance convergence. Our framework is implemented using PyTorch and trained on a single NVIDIA RTX 3090 GPU.

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TABLE I
DATASET PROPERTIES USED IN EXPERIMENTS

Properties	Train	Test					
Data Name	AID	AID	DIOR	DOTA	UCMerced	NWPU-RESISC45	RSSCN7
Used (Total) Image Numbers	3000 (10000)	900 (10000)	1000 (23463)	900 (2806)	1050 (1995)	900 (31500)	1050 (2800)
Select Number (per categories)	100	30	-	-	50	20	150
Original Size	600×600	600×600	800×800	800∼4000	256×256	256×256	400×400
Used Size	600×600	600×600	512×512	512×512	256×256	256×256	400×400
Spatial Resolution	0.5∼8m	0.5∼8m	-	-	0.3m	0.2∼30m	-
Categories	30	30	20	5	21	45	7
Task	Scene Classification	Scene Classification	Object Detection	Object Detection	Scene Classification	Classification	Classification

TABLE II
QUANTITATIVE COMPARISON ON DIOR [49], DOTA [50], AND AID [48] TEST SETS IN TERMS OF PSNR, SSIM, AND LPIPS. THE BEST, THE SECOND BEST, AND THE THIRD BEST RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY

Method Category	Method	Venue	DIOR			DOTA			AID		
			PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Base	Bicubic	-	28.45	0.7388	0.8377	31.54	0.7924	0.8520	29.07	0.7438	0.8624
CNN	EDSR	CVPRW'2017	30.74	0.8131	0.3101	34.35	0.8624	0.2578	31.19	0.8192	0.3084
	RDN	CVPR'2018	30.77	0.8137	0.3104	34.38	0.8629	0.2574	31.21	0.8197	0.3094
	FSRCNN	ECCV'2016	29.64	0.7813	0.3553	32.89	0.8301	0.3111	30.18	0.7870	0.3592
Attention	RCAN	ECCV'2018	30.86	0.8163	0.3055	34.49	0.8651	0.2532	31.29	0.8222	0.3031
	HAN	ECCV'2020	30.87	0.8166	0.3045	34.52	0.8654	0.2532	31.29	0.8223	0.3015
	NLSN	CVPR'2021	30.82	0.8152	0.3072	34.45	0.8643	0.2546	31.24	0.8209	0.3050
Transformer	SwinIR	ICCV'2021	30.83	0.8154	0.3084	34.52	0.8651	0.2543	31.29	0.8224	0.3029
	ART	ICLR'2023	30.88	0.8170	0.3061	34.53	0.8660	0.2522	31.30	0.8236	0.3004
	RGT	ICLR'2024	30.55	0.8110	0.3050	34.20	0.8611	0.2560	31.09	0.8175	0.3093
	TTST	TIP'2024	30.85	0.8159	0.3085	34.52	0.8657	0.2539	31.32	0.8234	0.3010
	FAT	TGRS'2025	30.74	0.8123	0.3116	34.23	0.8620	0.2592	31.09	0.8169	0.3135
Mamba	MambaIR	ECCV'2024	30.89	0.8174	0.3062	34.56	0.8663	0.2537	31.33	0.8243	0.2983
	FMSR	TMM'2024	30.71	0.8127	0.3034	34.28	0.8622	0.2527	31.19	0.8198	0.2994
	MaIR	CVPR'2025	30.90	0.8173	0.3057	34.57	0.8663	0.2536	31.34	0.8240	0.2994
	HAM	-	30.93	0.8180	0.3039	34.60	0.8669	0.2510	31.35	0.8240	0.2988
	HAM-Tiny	-	30.80	0.8153	0.3002	34.31	0.8643	0.2506	31.24	0.8216	0.2948

C. Datasets

In this article, we evaluate the SR performance of the proposed model on six widely used RSI benchmarks, namely, AID [48], DIOR [49], DOTA [50], UCMerced [51], RSSCN-7 [53], and NWPU-RESISC45 [52]. A detailed description of these datasets is provided in Table I. Following the methodology of prior work [20], AID is used for both training and testing. Specifically, we randomly select 3000 images for training and 900 images for testing, each with a resolution of 600 × 600. It is important to note that the training and testing subsets of AID are nonoverlapping. For model evaluation, we randomly extract 900 and 1000 images from DOTA and DIOR, respectively, both with a resolution of 512 × 512. In UCMerced, we utilize the predefined test set, which consists of 1050 images with a resolution of 256 × 256 pixels. For NWPU-RESISC45 and RSSCN7, we randomly sample 900 and 1050 images for testing, with resolutions of 256 × 256 and 400 × 400 pixels, respectively.

D. Evaluation Metrics

In image SR tasks, peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) [60] are the two most widely used image quality assessment (IQA) metrics. This article utilizes these two full-reference evaluation metrics to assess the SR reconstruction accuracy, with PSNR and SSIM calculated on the luminance channel (Y) in the YCbCr color space. In addition, to more comprehensively evaluate the perceptual quality of SR images, this article further adopts the learned perceptual image patch similarity (LPIPS) [61] metric.

E. Comparison With State-of-the-Art Methods

1) Comparative Methods: The SR methods compared in this article are categorized into four major types: CNN-based methods, attention-based methods, transformer-based methods, and Mamba-based methods. Specifically, the models include EDSR [7], RDN [8], FSRCNN [54], RCAN [9], HAN [10], NLSN [55], SwinIR [15], ART [16], RGT [56], TTST [33], FAT [57], MambaIR [19], FMSR [20], and MaIR [58].

2) Quantitative Evaluation: Tables II and III present detailed comparisons of reconstruction performance across all models on different datasets. The results show that our HAM achieves the best or near-best PSNR and SSIM scores across all datasets, clearly demonstrating its superior SR performance on various remote sensing benchmark datasets. Specifically, on the UCMerced dataset, HAM achieves a PSNR of 29.58 dB, surpassing the attention-based NLSN (29.45 dB) by

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TABLE III

QUANTITATIVE COMPARISON ON UCMERCED [51], RSSCN-7 [53], AND NWPU-RESISC45 [52] TEST SETS IN TERMS OF PSNR, SSIM, AND LPIPS. THE BEST, THE SECOND BEST, AND THE THIRD BEST RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY

Method Category	Method	Venue	UCMerced			RSSCN-7			NWPU-RESISC45		
			PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Base	Bicubic	-	26.91	0.7009	0.8602	27.98	0.6829	0.8683	27.64	0.6990	0.8365
CNN	EDSR	CVPRW'2017	29.29	0.7927	0.2887	29.32	0.7497	0.4307	29.41	0.7799	0.3376
	RDN	CVPR'2018	29.31	0.7924	0.2922	29.33	0.7501	0.4317	29.42	0.7805	0.3383
	FSRCNN	ECCV'2016	28.15	0.7521	0.3525	28.73	0.7236	0.4770	28.60	0.7466	0.3895
Attention	RCAN	ECCV'2018	29.44	0.7975	0.2797	29.38	0.7527	0.4252	29.49	0.7853	0.3305
	HAN	ECCV'2020	29.43	0.7975	0.2777	29.37	0.7525	0.4216	29.49	0.7833	0.3300
	NLSN	CVPR'2021	29.45	0.7974	0.2840	29.37	0.7520	0.4285	29.47	0.7824	0.3330
Transformer	SwinIR	ICCV'2021	29.47	0.7980	0.2819	29.38	0.7531	0.4239	29.49	0.7832	0.3334
	ART	ICLR'2023	29.50	0.8002	0.2788	29.42	0.7546	0.4214	29.52	0.7848	0.3292
	RGT	ICLR'2024	29.27	0.7917	0.3035	29.34	0.7503	0.4302	29.34	0.7774	0.3384
	TTST	TIP'2024	29.48	0.7989	0.2800	29.39	0.7537	0.4215	29.51	0.7839	0.3328
	FAT	TGRS'2025	29.36	0.7950	0.2845	29.33	0.7497	0.4320	29.42	0.7801	0.3354
Mamba	MambaIR	ECCV'2024	29.47	0.7989	0.2778	29.40	0.7544	0.4203	29.53	0.7852	0.3300
	FMSR	TMM'2024	29.37	0.7949	0.2836	29.34	0.7515	0.4178	29.42	0.7812	0.3295
	MaIR	CVPR'2025	29.51	0.7994	0.2793	29.42	0.7547	0.4199	29.53	0.7853	0.3285
	HAM	-	29.58	0.8016	0.2775	29.44	0.7549	0.4196	29.55	0.7854	0.3297
	HAM-Tiny	-	29.45	0.7978	0.2831	29.37	0.7526	0.4168	29.50	0.7833	0.3317

0.13 dB, the transformer-based ART (29.50 dB) by 0.08 dB, the Mamba-based MambaIR (29.47 dB) by 0.11 dB, and MaIR (29.51 dB) by 0.07 dB. In addition, in perceptual quality assessment, HAM attains the best LPIPS scores of 0.2510 and 0.2775 on the DOTA and UCMerced datasets, respectively, further validating its advantage in visual perceptual quality.

According to the results in Tables II and III, we can further find that the overall performance of the existing Mamba-based methods surpasses that of the transformer-based and CNN-based methods. However, there are also instances where the opposite occurs on some datasets. For example, MambaIR [19] performs worse than the transformer-based ART [16] on datasets UCMerced and RSSCN-7. Nevertheless, our proposed HAM has achieved the consistently best results on all six test datasets, demonstrating the stronger generalization ability of our method.

Table IV presents a complexity comparison of various methods in terms of parameters, FLOPs, inference time, and memory consumption. Our analysis shows that the HAM method outperforms the Mamba-based FMSR across all complexity metrics and reduces inference time by 21.123 ms compared to the transformer-based ART. Moreover, HAM achieves higher PSNR values in reconstructed images, demonstrating a superior balance between reconstruction quality and computational efficiency. In addition, as shown in Tables II–IV, our lightweight model, HAM-Tiny, significantly reduces complexity while maintaining comparable reconstruction performance, underscoring its strong competitive advantage.

3) Qualitative Results: To evaluate the visual quality of the HAM image reconstruction results, we conducted a detailed comparison with several classical models, including EDSR [7], HAN [10], NLSN [55], SwinIR [15], ART [16], MambaIR [19], and FMSR [20]. We provided a diverse set of test scenarios for the model using six test datasets, which facilitates a comprehensive assessment of the model's robustness and recovery performance under various conditions.

TABLE IV

COMPARISON OF DIFFERENT METHODS ON PARAMETERS, FLOPS, INFERENCE TIME, AND MEMORY. THE FLOPS RESULTS ARE CALCULATED WITH AN INPUT TENSOR OF SIZE $1 \times 3 \times 600 \times 600$. THE INFERENCE TIMES ARE TESTED ON ONE RANDOM IMAGE. PSNR IS THE RESULT OF THE UCMERCED DATASET. THE TRAIN TIME IS THE TIMING AFTER 500 000 TRAINING ITERATIONS ON A SINGLE NVIDIA 3090

Method	Params (M)	FLOPs (G)	Time (ms)	Memory (MB)	train (h)	PSNR (dB)
NLSN	44.156	1218.99	157.157	181.458	45	29.94
HAN	64.198	1475.76	209.838	257.50	46	29.92
ART	16.545	610.65	359.508	75.94	98	30.01
RGT	13.365	354.11	447.830	186.46	68	29.76
MambaIR	20.569	620.68	294.734	91.20	114	29.96
MaIR	19.592	595.46	313.711	87.920	117	30.00
FMSR	36.931	831.33	405.878	153.79	100	29.91
HAM	25.195	667.86	338.385	109.01	118	30.09
HAM-Tiny	8.088	255.19	187.038	43.56	76	29.95

The visual comparison results for the AID, DIOR, and DOTA datasets are displayed in Fig. 7, focusing on 4× SR. Through visual analysis, we observe that HAM excels in restoring details, edge sharpness, and texture, particularly in capturing key high-frequency details in RSIs. For instance, in the "playground_136" image presented in Fig. 7, methods based on CNNs, attention mechanisms, the transformer-based ART model, and the Mamba-based FMSR method all produced noticeable artifacts. In contrast, HAM's recovery in the lower-right corner of the playground is closer to the HR image than those of SwinIR and MambaIR, with the runway lines appearing sharper.

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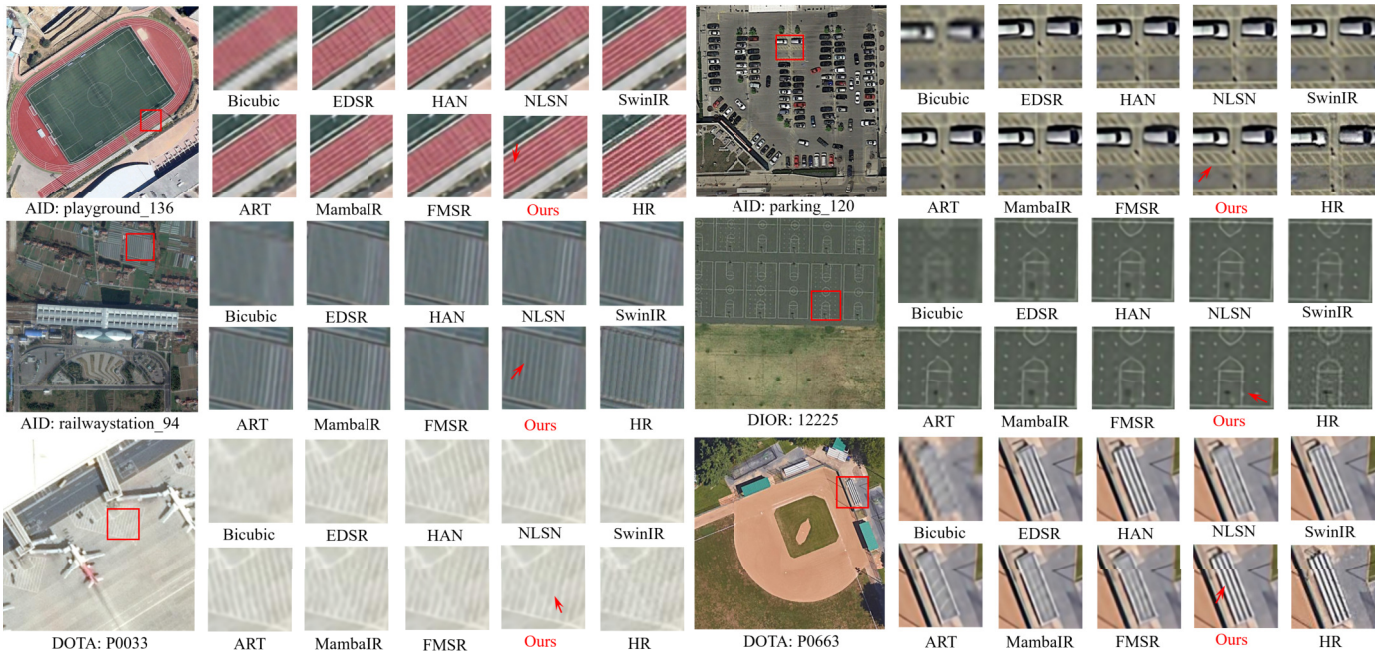


Fig. 7. Visual comparison of HAM on the AID, DIOR, and DOTA datasets with CNN-based, attention-based, transformer-based, and Mamba-based methods. The details are magnified for better observation and comparison.

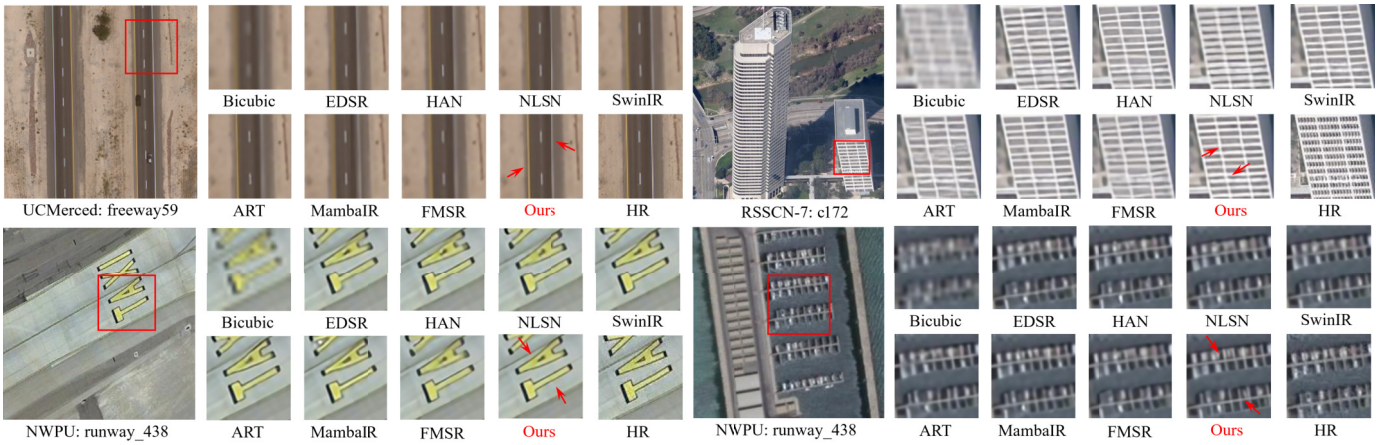


Fig. 8. Visual comparison of HAM on the UCMerced, RSSCN-7, and NWPU-RESISC45 datasets with CNN-based, attention-based, transformer-based, and Mamba-based methods. The details are magnified for better observation and comparison.

In addition, in the "railwaystation_94" image, the farmland furrows exhibit a regular striped pattern. Except for SwinIR and HAM, the other models failed to fully recover the stripe details of the land. Among them, the stripes recovered by HAM are clearer, while MambaIR, although able to restore most stripe details, still exhibits artifacts in the upper portion. This indicates that feature modeling solely in the spatial domain may not suffice for fully recovering the complex details of RSIs, whereas combining spatial and frequency domain feature learning enhances the accurate reconstruction of high-frequency information.

The visual comparison results for the UCMerced, RSSCN-7, and NWPU-RESISC45 datasets are shown in Fig. 8. This includes the SR reconstruction results for the "runway_438" image. We observed that EDSR, HAN, NLSN, ART, and FMSR exhibited blurriness or unclear recovery when reconstructing the top line of the letter "A" and the tail and bottom lines of the letter "T." MambaIR, in contrast, reconstructed the lines of the letter "T" too thinly, resulting in a significant difference from the HR reference image. In comparison, the edges of the lines restored by HAM were smoother, and the overall shape more closely matched the HR reference image, demonstrating superior detail preservation. In the recovery process of "freeway59" in Fig. 8, all models successfully reconstructed most of the road structure. However, when it came to restoring the yellow and white line details on both sides of the road, most models either restored only one side of the markings or rendered both sides blurry. In contrast, HAM was able to restore the edge details of the yellow and white lines more clearly,

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TABLE V
ABLATION STUDY ON DIFFERENT COMPONENTS. CBCA MEANS THE CONVBLOCKS AND CA IN MAMBAIR [19]

Model	HAA	CBCA	HSFA	DFSE	Params (M)	UCMerced		DOTA	
						PSNR	SSIM	PSNR	SSIM
(a)		✓			20.569	29.47	0.7989	34.56	0.8663
(b)	✓	✓			22.902	29.52	0.7998	34.60	0.8667
(c)			✓		15.546	29.51	0.7997	34.58	0.8664
(d)	✓		✓		17.879	29.55	0.8007	34.58	0.8667
(e)			✓	✓	22.862	29.54	0.8011	34.59	0.8668
(f)	✓		✓	✓	25.195	29.58	0.8016	34.60	0.8669

TABLE VI
ABLATION STUDY ON FOURIER LOSS

Loss	AID		DIOR	
	PSNR	SSIM	PSNR	SSIM
L1 loss	31.3537	0.8240	30.9345	0.8180
Fourier Loss (lr=2 $\times$ 10^{-4})	31.3526	0.8239	30.9094	0.8176
Fourier Loss (lr=1 $\times$ 10^{-4})	31.3348	0.8234	30.8795	0.8168
Fourier Loss (lr=2 $\times$ 10^{-4}) + Mixup	31.3439	0.8241	30.9029	0.8178

TABLE VII
ABLATION STUDY ON HAA AND LEARNING RATE

Block	DIOR		DOTA	
	PSNR	SSIM	PSNR	SSIM
Baseline (MambaIR)	30.8946	0.8174	34.5563	0.8663
add HAA (lr=2 $\times$ 10^{-4})	30.9261	0.8178	34.6003	0.8667
add HAA (lr=1 $\times$ 10^{-4})	30.8990	0.8167	34.5841	0.8660

further highlighting its advantage in recovering high-frequency information.

F. Ablation Studies

In this section, we conduct a comprehensive analysis of the effectiveness of HAM by examining its main structure and its variants. All models are trained and tested on the same datasets described in Section IV-C, with 500 000 iterations performed on the AID dataset.

1) Ablation Study on Overall Structure: To evaluate the components of HAM, we conducted ablation experiments using six different configurations, with results presented in Table V. The models are defined as follows.

Model (a): Uses MambaIR as the baseline comparison model.

Model (b): Incorporating the HAA into the baseline model improves accuracy on the UCMerced and DOTA datasets, demonstrating the effectiveness of multiscale learning across layers.

Model (c): In this configuration, the convolutional blocks and CA (CBCA) in the baseline model are replaced with the HSFA as the spatial domain learning module. Compared to the baseline, our proposed HSFA employs a dual-branch structure to separately learn spatial information and channel features, effectively addressing the issues of local information loss and channel redundancy present in MambaIR. On the UCMerced and DOTA datasets, the PSNR increases by 0.04 and 0.02 dB, respectively, while the number of parameters decreases by 5.023 M. This indicates that HSFA enhances feature representation capability while maintaining high computational efficiency.

Model (d): Building upon the introduction of HAA, and the HSFA replacing the original CBCA in the baseline. Compared to Model (c), which employs only HSFA, Model (d) improves the PSNR by 0.04 dB and 0.01 dB on the UCMerced and DOTA datasets, respectively. These results further underscore the advantages of HAA and emphasize the importance of exploring interlayer information interactions.

Model (e): In this configuration, the CBCA in the baseline are replaced with dual-domain learning, enabling the model to learn the land-cover features of RSIs from both spatial and frequency domains. Compared to Model (c), which utilizes only spatial domain learning, this approach increases the PSNR by 0.03 and 0.01 dB on the UCMerced and DOTA datasets, respectively. This indicates that incorporating frequency-domain information enhances the recovery of richer high-frequency details.

Model (f): For our proposed HAM model, which integrates HAA, HSFA, and DFSE dual-domain learning, performance is optimized as indicated in Table V, further validating the effectiveness of HAM in the task of RSISR.

2) Study on Fourier Loss and Learning Rate: To assess the impact of Fourier Loss and Mixup regularization on model performance, we conducted ablation experiments comparing the effects of different configurations. The results are shown in Table VI. We found that adding Fourier Loss and Mixup regularization to the baseline model did not result in significant improvements in PSNR and SSIM, and in some datasets, there was even a slight decrease. This indicates that, although Fourier Loss emphasizes frequency-domain information and Mixup regularization enhances the model's generalization ability, their contributions to the current HAM architecture are relatively limited. Therefore, we ultimately decided to retain the baseline model's L1 loss function and remove Fourier Loss and Mixup regularization to ensure the model's simplicity and stability.

3) Study on HAA and Learning Rate: We investigated the impact of introducing the HAA module and adjusting the initial learning rate on the model performance, using the baseline MambaIR model. The experimental results are shown in Table VII. The data indicate that the addition of the HAA module results in an improvement in model performance. Specifically, for the DIOR dataset, with learning rates of 2×10^{-4} and 1×10^{-4}, the PSNR increased by 0.0315 and 0.0044 dB, respectively, compared to the baseline. In the case of the DOTA dataset, the introduction of HAA yielded more significant improvements, with the PSNR gains by 0.044 and 0.0278 dB at learning rates of 2×10^{-4} and 1×10^{-4}, respectively.

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TABLE VIII
ABLATION STUDY ON HAA AND OTHER ATTENTIONS

Block	Params(M)	UCMerced		DOTA	
		PSNR	SSIM	PSNR	SSIM
Baseline (HAM)	25.195	29.5778	0.8016	34.6029	0.8669
w/o HAA	22.862	29.5440	0.8016	34.5879	0.8668
w SElayer	25.340	29.5515	0.8004	34.5381	0.8661
w ECA	25.196	29.5027	0.7988	34.4761	0.8653

TABLE IX
ABLATION STUDY ON DFSE FOR FREQUENCY MODULE EFFECTIVENESS

Block	DOTA		UCMerced	
	PSNR	SSIM	PSNR	SSIM
Baseline (HAM)	34.6029	0.8669	29.5778	0.8016
w/o DFSE	34.5840	0.8667	29.5471	0.8007
w FSM	34.3178	0.8622	29.4233	0.7961

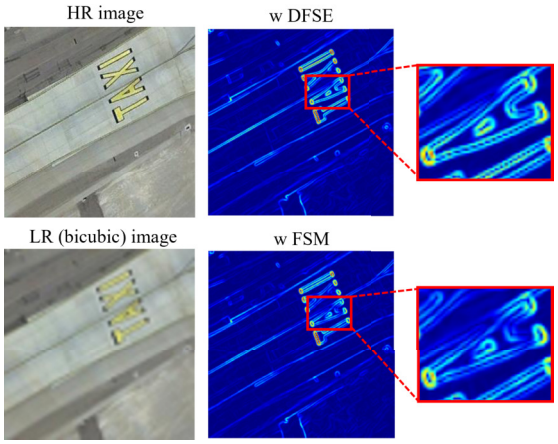


Fig. 9. Comparison of spatial structure reconstruction: DFSE versus FSM. The proposed DFSE module produces enhanced edge details compared to the frequency selection module (FSM).

Furthermore, we conduct ablation experiments by replacing the proposed HAA mechanism with alternative attention modules, including the Squeeze-and-Excitation (SE) module [46] and the Efficient Channel Attention (ECA) module [62]. Since HAA directly operates on 4-D interlayer features ($N \times C \times H \times W$), whereas SE and ECA are designed for 3-D inputs, we reshape the feature maps into a compatible form ($(NC) \times H \times W$) prior to processing. As shown in Table VIII, HAA achieves higher PSNR than SE and ECA by 0.0263 and 0.0751 dB, respectively, on the UCMerced dataset. These results demonstrate the effectiveness of HAA in enhancing reconstruction quality. Specifically, the incorporation of the difference matrix D within HAA enables more distinct and efficient multilevel feature integration, which directly contributes to its superior performance.

4) Study on DFSE: To evaluate the effectiveness of the proposed frequency module, we conduct experiments by removing the DFSE module and replacing it with the FSM [20]. As shown in Table IX, the DFSE module achieves PSNR improvements of 0.2851 and 0.1545 dB over the FSM on the DOTA and UCMerced datasets, respectively. This demonstrates that the proposed amplitude–phase decoupling followed by dedicated CA facilitates more effective utilization of frequency-domain information for targeted feature learning. Moreover, the performance degradation caused by FSM suggests that merely introducing convolutional operations in the Fourier domain is suboptimal for effective frequency modeling. Compared to the baseline without any frequency module, these results also indicate that FSM may not be suitable for serial integration within our architecture. As illustrated in Fig. 9, visual comparisons of spatial structures reveal that DFSE produces sharper edge restoration and more complete reconstruction of the letter "A," whereas FSM yields noticeably blurrier contours.

V. CONCLUSION

This article proposes the HAM network for RSISR. HAM addresses limitations of prior Mamba-based models in hierarchical feature interaction and local–global feature extraction. It introduces two key modules: the HAA for enhanced multilevel feature integration, and the SFIIM, which incorporates frequency-domain enhancement and spatial dual-branch aggregation. Extensive experiments on six benchmark datasets demonstrate that HAM achieves superior reconstruction performance efficiently. These findings highlight HAM's potential as a strong, scalable, and efficient framework for advancing RSISR.

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